Software Requirements Specification

for

Pawgress

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Lib Erenea Nuevo, Hannah Bacomo, Jacob Tan, Shaan Dayag	January 23, 2026	• Filled sections 1 to 4	1
Lib Erenea Nuevo, Hannah Bacomo, Jacob Tan, Shaan Dayag	February 13, 2026	• Added Iteration Plan • Added Gantt Chart	2
Lib Erenea Nuevo, Hannah Bacomo, Jacob Tan, Shaan Dayag	February 27, 2026	• Adjusted Gantt chart distribution • Edited iteration plans to match Gantt chart • Added User Case Diagram and Class Diagram	3

1. Introduction

1.1 Purpose

This document contains all pertinent details and functionalities of Pawgress version 1.0 — an app that gamifies productivity, incentivizing users to create consistent habits by having them take care of a chosen pet. The app is primarily for the student body of the Ateneo de Manila University. This document defines the functional and non-functional requirements, user interfaces, and constraints of the development team, Yvonne Adarna, to guide the development, testing, and deployment of the system.

1.2 Document Conventions

For the purposes of this document, listed below are the meanings of the terminology and acronyms used further in the document:

- Pawgress – The name of the application.
- Streak – A measurement of consecutive days a user completes their habits or tasks.
- AdMU – Ateneo de Manila University
- SRS – Software Requirements Specification
- XP – Experience Points

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification document is intended for software developers who will develop Pawgress and future contributors who may enhance or maintain the system, as well as individuals who are required to read this document in fulfillment of academic requirements.

The SRS document is composed of six main parts. The first part presents the introduction and background of the product. This is followed by the overall description of Pawgress, which includes the product perspective, operating environment, user classes, and documentation. The third part discusses the external interface requirements such as user interfaces, hardware interfaces, and software interfaces. The fourth part elaborates on the system features and core functionalities of the application. The fifth part covers the non-functional requirements, including performance, security, safety, business rules, and software quality attributes. Lastly, the document includes appendices that provide additional information, such as the glossary, analysis models, team organization, and items to be determined.

It is recommended that developers, testers, and stakeholders read each section thoroughly to ensure a complete understanding of the system requirements and project scope.

1.4 Product Scope

Pawgress is a web-based productivity and journaling application designed to help students develop consistent habits and improve time management through incentivizing task completion and study sessions. The system integrates productivity features, allowing the user to gain XP and currency. Gaining more XP allows the user to move to the next level, where they can unlock more pets or accessories. Currency can be used to customize the user's pet, such as by purchasing accessories. Users can also add friends to view their study statistics and pet avatars.

Studying may seem like a daunting task for many students. Many current productivity websites and applications fail to resolve this issue because they simply focus on the studying aspect without appealing to the happiness and enjoyment of their users. To address this, Pawgress aims to combine productivity and entertainment in order to motivate students and turn studying into a more enjoyable experience.

1.5 References

The group will make use of Figma for designing and creating the interface mockups and prototypes of the Pawgress application. The official Figma design resources will serve as references for the interface layouts and component designs. Online design pegs and visual inspiration material platforms such as Pinterest will be consulted to guide the application's visual style, structure, and overall user interface aesthetics. In addition to these, technical documentation and resources such as React.js, Bootstrap, and Django documentation, along with platforms such as W3Schools, will be used to support the development of the program.

2. Overall Description

2.1 Product Perspective

The name Pawgress is a combination of "paw," which represents the unique virtual pet care feature of the application, and "progress," which reflects the sense of completion and improvement that the app wishes to convey to its users. At present, students often rely on separate tools to manage their academic responsibilities. In addition, maintaining motivation and consistency remains a challenge due to the repetitive nature of productivity tasks. As a result, users are required to switch between different platforms while struggling to sustain long-term engagement.

This app addresses these limitations by integrating both common productivity app features — to-do lists, calendaring, Pomodoro timers — as well as features from games — customization, leveling and experience points, shops and currency — to make studying both organized and enjoyable.

2.2 Product Functions

- User Registration and Authentication: Account creation, login, and secure access control for buyers and sellers.
- To-do List: Enables users to add, edit, delete, and organize tasks, set deadlines, and select the status of tasks (i.e., complete, in progress, not yet completed)
- Gamification and Virtual Pet Reward System: A reward-based engagement system where users earn XP and virtual currency by completing platform activities such as successful transactions, posting verified listings, leaving reviews, and maintaining activity streaks. Earned currency is used to care for a virtual pet whose status (such as mood, health, and cleanliness) reflects user engagement. This feature encourages consistent participation and responsible platform usage.
- Socialization: Users can add friends on the website and look at their profiles and pets
- Pomodoro Timer: Provides built-in focus timers with pre-defined work and break intervals that users can choose from to help them manage study sessions and improve concentration.
- Calendar System: Provides users a weekly and monthly overview of their tasks, deadlines, and study sessions based on their to-do lists and activity data.

2.3 Users Classes and Characteristics

The main users of Pawgress are students, likely ranging from secondary to tertiary level. These users are technologically proficient and often use apps and websites on their cellphones, laptops, or computers. They may also face challenges with studying or task management, given the higher and more difficult level of their education or extracurricular activities. Given these factors, they may be looking for an app or website that could motivate them to consistently be productive.

2.4 Operating Environment

The host computer that will run the server must be able to run Windows 8, Windows 8.1, Windows 10, macOS Mojave, macOS Catalina, macOS Big Sur, and Python 3. The application will operate as a web-based platform accessible via modern browsers (Chrome, Firefox, Safari, etc.). Development and testing will be conducted on Windows/macOS laptops and mobile devices.

2.5 Design and Implementation Constraints

The design and implementation of Pawgress will be created using only free and/or open-source software and services, and the hardware already available to the developers. The application will be implemented as a web-based system using Java and JavaScript as the programming languages to be used, along with React.js for front-end development and Bootstrap for interactive CSS styling. Figma will be used to create mockups of the website.

These technology choices may impose limitations on the system architecture, performance optimization strategies, and design patterns that may be applied. The system will be designed to operate using standard web communication protocols such as HTTP and HTTPS to support secure data transmission and client-server communication.

2.6 User Documentation

A tutorial will be made to assist new users of the website in the form of a 'Getting Started' page that will guide users on how to utilize the different functionalities of the site. A README file or "Getting Started" guide will be provided to explain the reward mechanics and productivity tracking integration on the site. Beyond that, a 'Frequently-Asked Questions' page may also be implemented to address common user concerns and questions.

2.7 Assumptions and Dependencies

The development of Pawgress assumes that users will have a modern browser that supports current web technologies such as JavaScript and HTML5. It is also assumed that resources such as React.js and Bootstrap will remain compatible with and supported throughout the development and testing period. Any major changes in these technologies may affect the product's implementation and maintenance.

3. External Interface Requirements

3.1 User Interfaces

At the time of writing, the user interface of Pawgress has not yet been finalized. However, the application is expected to include basic interface components such as a navigation bar, buttons, dashboard views, and user profiles.

Detailed interface design will be documented during the design phase using mockups created in Figma.

3.2 Hardware Interfaces

Pawgress will run on standard consumer hardware to ensure functional and responsive operation. The system supports modern input and output devices without requiring specialized components.

- Supported Devices: Personal computers and laptops.
- Input Methods: Keyboard and mouse.
- Output: Standard display screens.
- Browsers: Compatible with modern web browsers (Chrome, Firefox, Safari, etc.).
  Hardware Requirements: No specialized hardware needed for the application to function.

3.3 Software Interfaces

Pawgress will interact with several software components to function properly:

- Front-End: Uses React.js for dynamic UI and Bootstrap for responsive design.
- Back-End: Implemented in Java and JavaScript, handling business logic and user requests.

- Database: Stores user accounts, habits, to-do lists, virtual pet stats, XP/currency, friends, and calendar events.
- Development Tools: Figma for interface mockups and design reference.
- Communication: Exchanges data with the client over HTTP/HTTPS. User actions are validated, stored, and returned to update the UI.

All libraries and tools used are free or open-source, ensuring accessibility for the development team.

3.4 Communication Interfaces

The application shall utilize HTTP/HTTPS protocols for data transmission between the client-side browser and any potential backend services. It will be accessible through modern web browsers that support JavaScript and the React.js library. A stable internet connection is required for users to update pet statistics and manage account data in real-time.

4. System Features

4.1 Virtual Pet Care System

4.1.1 Description and Priority

This feature allows users to select and take care of a pet. The user can name this pet and customize its initial appearance. The user can interact with the pet by feeding it, playing with it, and dressing it up. This feature is of high priority.

4.1.2 Stimulus/Response Sequences

To utilize this feature, the user will need to have created an account on the website and selected a pet. They must have finished a tutorial sequence where they name the pet and choose its initial appearance. After that, the pet can be interacted with through the home screen by clicking on the different buttons such as “Feed,” “Play,” or “Dress Up.”

4.1.3 Functional Requirements

REQ-1: The user will need to be logged in; otherwise, the pet cannot be displayed, as it will prompt the user to log in.

REQ-2: The user will need to finish the tutorial sequence, as they cannot proceed without having done so.

REQ-3: The user can only feed the pet if their hunger bar is not full; otherwise, instead of proceeding, a message will pop up indicating that the pet is full.

4.2 To-do List Maker

4.2.1 Description and Priority

This feature allows users to create and customize tasks in their to-do lists. In task creation, users can name the task, set a deadline, set a difficulty (easy, medium, hard), and select a category (which are also created by the users, e.g. Academics, Organization, Personal). Users can also create subtasks under each task. Users can customize their to-do list by adding, deleting, or marking tasks as done. This feature is of high priority.

4.2.2 Stimulus/Response Sequences

In order to utilize this feature, the user must be logged in. The user must go to the “Productivity” section of the website and navigate to the “To-do List.” The users can click the “Create Task”, “Edit”, or “Delete” button.

4.2.3 Functional Requirements

REQ-1: The user will need to be logged in; otherwise, they will be prompted to log in.

REQ-2: When creating a task, the user cannot leave the task name empty, or else an error message will pop up.

REQ-3: If the user’s list is empty, they will not be able to edit the to-do list. The edit button is only available if there is at least one task.

4.3 Pomodoro Timer

4.3.1 Description and Priority

The Pomodoro Timer is a time-management tool designed to help users maintain focus by breaking work into intervals separated by short breaks. Users can customize the intervals and have the option to play background music while studying. In Pawgress, this feature serves as a primary method for users to earn virtual currency and XP. This is a high-priority feature as it directly facilitates the core game loop of the application.

4.3.2 Stimulus/Response Sequences

The user selects a work interval duration and begins the timer. The system initiates the countdown and displays the remaining time. After the work interval timer reaches zero, the system triggers a notification, prompting the user to start a break, and awards the user virtual currency and XP. If the user chooses to “cancel” before the timer finishes, the system halts the countdown and resets the timer. No virtual currency or XP is awarded for the incomplete session.

4.3.3 Functional Requirements

REQ-1: The system should allow users to start, pause, and reset the countdown timer.

REQ-2: The users are allowed to customize the duration of focus intervals and breaks through a settings menu.

REQ-3: When focus or break intervals are completed, the system will trigger a distinct auditory and visual notification.

REQ-4: The user must be awarded a specific amount of virtual currency and XP only upon the successful completion of a focus interval.

REQ-5: These rewards must scale with the length of the focus interval.

REQ-6: The system will automatically track and display the total number of focus sessions completed by the user in the current day.

4.4 Calendar Manager

4.4.1 Description and Priority

This feature allows the user to manage a personal calendar. Users can create and delete events on each calendar day. Users can also view the details of each event, such as the time, category, etc. This feature is of medium priority.

4.4.2 Stimulus/Response Sequences

The user must be logged in to manage their calendar. The user must go to the “Productivity” section of the website and navigate to the “Calendar.” Here, users can click the “Add” button to add events, and can click on each event entry to view more details.

4.4.3 Functional Requirements

REQ-1: The user should be logged in and on the Calendar page.

REQ-2: The user must click on the Add button to populate the calendar.

4.5 Friending System

4.5.1 Description and Priority

This feature allows a user to add other friends and view their profiles, statistics, and pets. This feature is of medium priority.

4.5.2 Stimulus/Response Sequences

The user must be logged in to see and add friends. The user must navigate to the “Friends” page and input a username. Afterwards, there will be a button to send a friend request. Users can accept or decline friend requests they have received. Upon accepting a friend request, that user will appear on a friend list, and clicking on them will show their profile, including their pet and study statistics (e.g., streak of how many days they have studied, currency earned, level, etc.).

4.5.3 Functional Requirements

REQ-1: The user should be logged in and on the Friends page.

REQ-2: The user must input an existing username; otherwise, a message containing the text “No users were found” will be displayed.

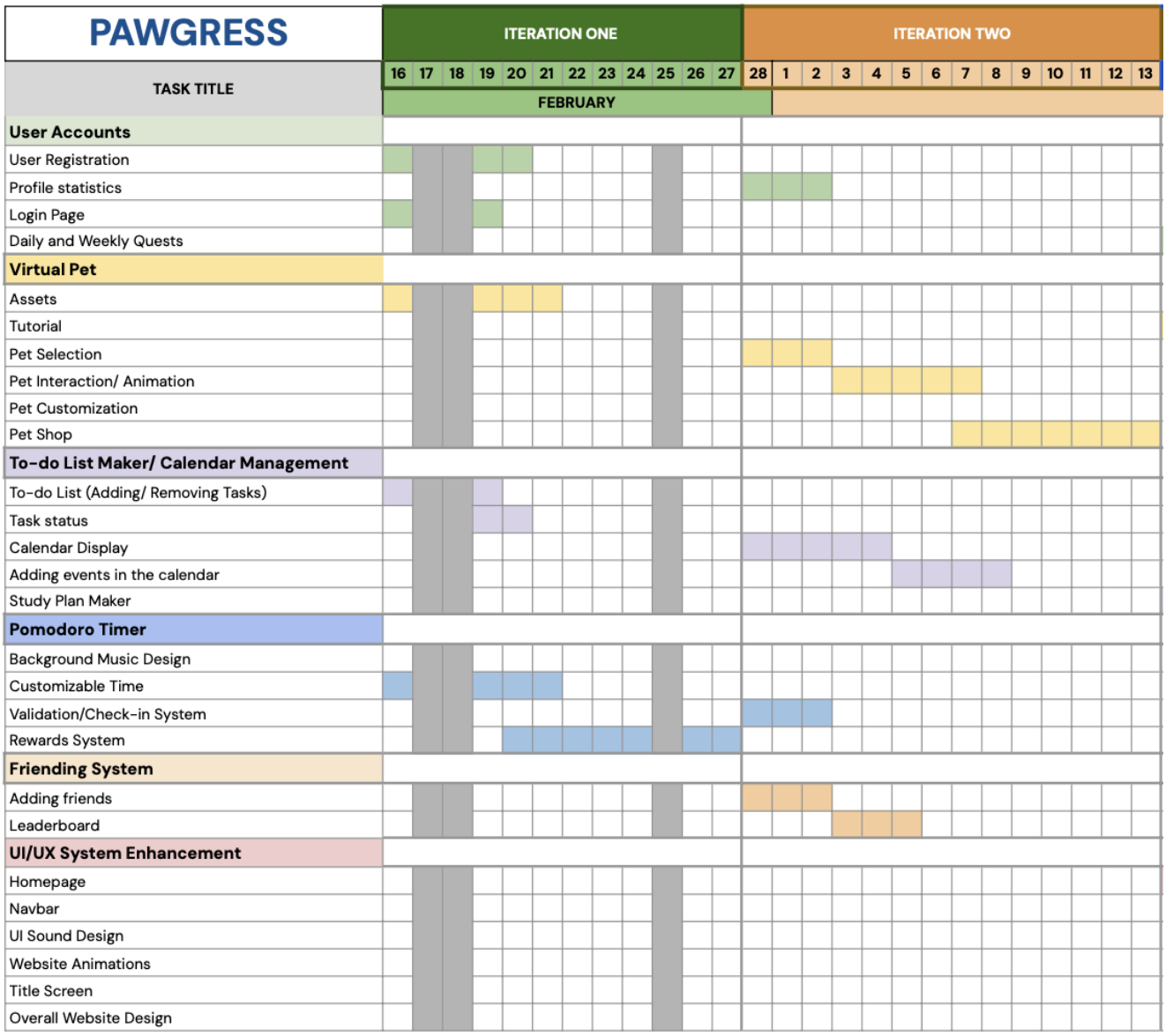
5. Iteration Plan

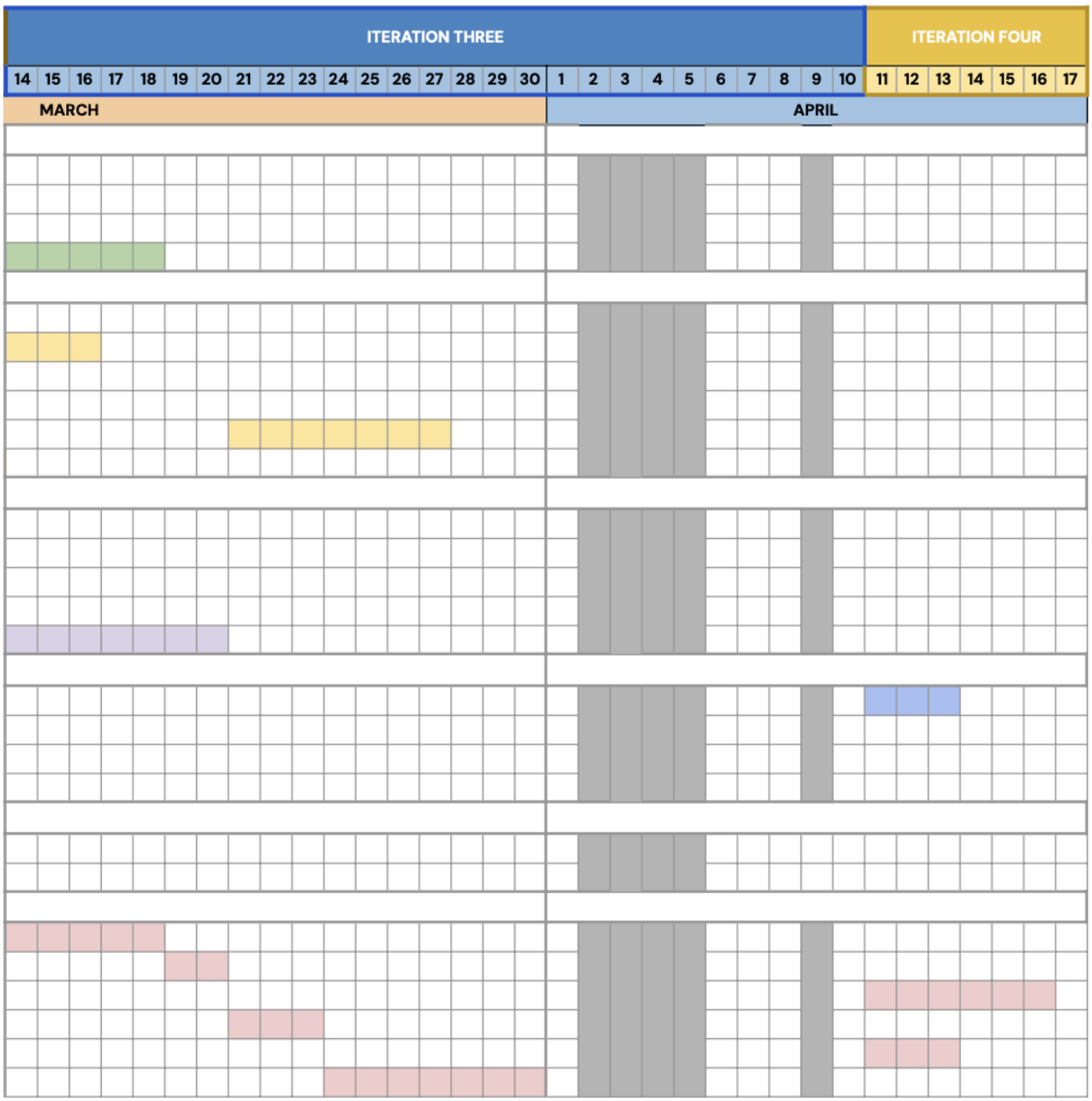
5.1 Task Assignments

Name/ Description	Priority	Iteration	Assigned To	Point Estimate
User Registration	High Priority	1st	Eren Nuevo	2
Login Page	High Priority	1st	Eren Nuevo	2
Pet Assets	High Priority	1st	Louise Angelo Tenorio	3
To-do List	High Priority	1st	Hannah Denise Bacomo	3
Task Status	High Priority	1st	Hannah Denise Bacomo	3
Customizable Pomodoro Timer	High Priority	1st	Jacob Tan	3
Scaling Reward System from Pomodoro	High Priority	1st	Jacob Tan, Shaan Graal Dayag	3
Profile Statistics	Low Priority	2nd	Shaan Graal Dayag	2
Pet Selection	High Priority	2nd	Shaan Graal	3

Process			Dayag	
Pet Interactions and Animations	Medium Priority	2nd	Louise Angelo Tenorio	3
Pet Shop	High Priority	2nd	Eren Nuevo	5
Calendar Display	Medium Priority	2nd	Hannah Denise Bacomo	3
Adding events in calendar	Medium Priority	2nd	Hannah Denise Bacomo	3
Leaderboard	Low Priority	2nd	Shaan Graal Dayag	2
Pomodoro Check-in System	Low Priority	2nd	Jacob Tan	2
Friending System	Medium Priority	2nd	Jacob Tan	3
Leaderboard	Low Priority	2nd	Shaan Graal Dayag	2
Daily and Weekly Quests	Low Priority	3rd	Hannah Denise B. Bacomo	3
Tutorial	Low Priority	3rd	Louise Angelo Tenorio	3
Pet Customization	High Priority	3rd	Eren Nuevo	3
Study Plan Maker	Medium Priority	3rd	Louise Angelo Tenorio	5
Homepage UI/UX Enhancement	High Priority	3rd	Louise Angelo Tenorio	2
Navigation Bar	High Priority	3rd	Hannah Denise Bacomo	1
Website Animations	Medium Priority	3rd	Jacob Tan	3
Overall Website Design	High priority	3rd	Eren Nuevo	3
Pomodoro BGM Design	Low Priority	4th	Jacob Tan	
UI Sound Design	Low Priority	4th	Jacob Tan	1
Title Screen	Low Priority	4th	Eren Nuevo	2

5.2 Gantt Chart





6. Other Nonfunctional Requirements

6.1 Performance Requirements

<If there are performance requirements for the product under various circumstances, state them here and explain their rationale, to help the developers understand the intent and make suitable design choices. Specify the timing relationships for real time systems. Make such requirements as specific as possible. You may need to state performance requirements for individual functional requirements or features.>

TBD

Pet animations and status updates must be reflected within 1 second after user interaction. There should be no lag or stuttering when playing music or audio when using the Pomodoro timer.

6.2 Safety Requirements

<Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied.>
TBD

Sensitive information from users' profiles must be encrypted.

6.3 Security Requirements

<Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements. Refer to any external policies or regulations containing security issues that affect the product. Define any security or privacy certifications that must be satisfied.>
TBD

6.4 Software Quality Attributes

<Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.>
TBD

The pet care system, to-do list maker, Pomodoro timer, calendar manager, and pet care system are all supported by both mobile and desktop web browsers without essential features being unusable. Pet animations and status updates must be reflected within 1 second after user interaction. There should be no lag or stuttering when playing music or audio when using the Pomodoro timer. Adding, deleting, or editing a task must take no more than 1 second for the change to take effect. The calendar interface should not look cluttered, even if there are many task entries.The system is capable of storing at least 50 users.

6.5 Business Rules

<List any operating principles about the product, such as which individuals or roles can perform which functions under specific circumstances. These are not functional requirements in themselves, but they may imply certain functional requirements to enforce the rules.>
TBD

7. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>
TBD

The app should not allow or promote actions that could be interpreted as animal abuse.

UML Diagrams

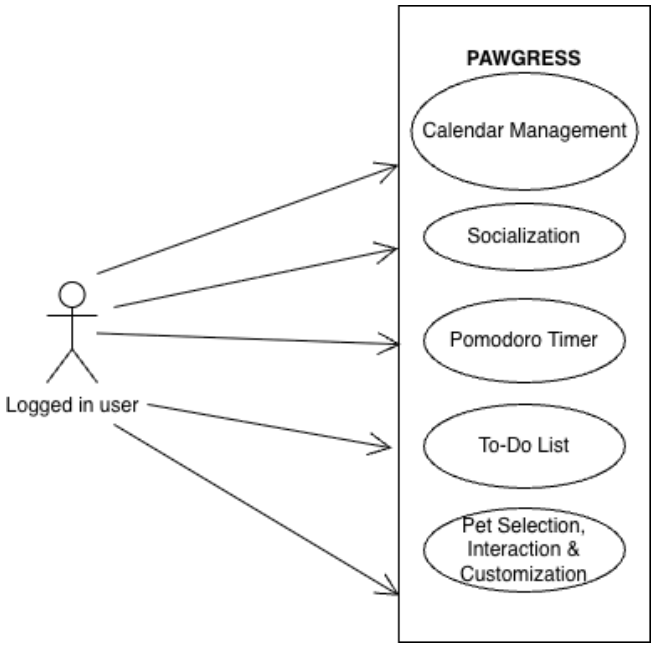


Figure 1. Use-Case Diagram – Pawgress

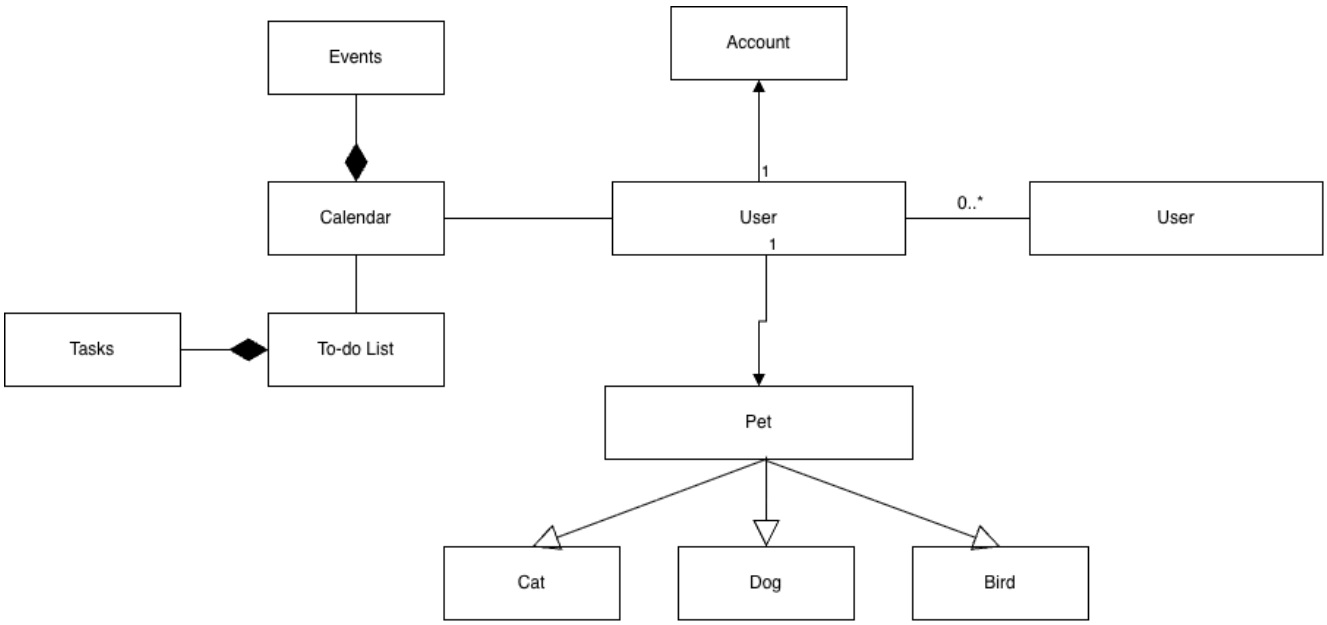


Figure 2. Class Diagram – Pawgress

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>
TBD

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>
TBD

Appendix C: Team Organization and Roles

The Pawgress will be using Facebook Messenger to coordinate. The team will be following the democratic structure in working on the project. The roles of the team members are as follows:

- Hannah Denise Bacomo - Main developer for features 4.2 and Designer
- Shaan Graal Dayag - Main developer for feature 4.4 and Designer/ Audio Designer
- Lib Erenea Nuevo - Main developer for feature 4.1 and Designer
- Jacob Tan - Main developer for feature 4.3 and Audio Designer

- Louise Angelo Tenorio - Main developer for feature 4.5

Appendix D: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>
TBD